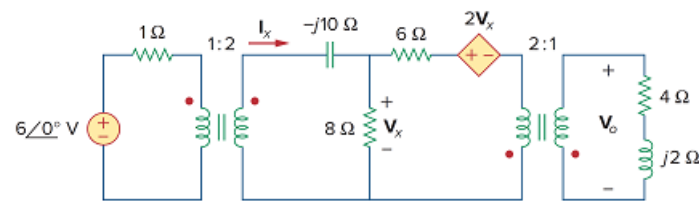


Problem

Find I_x and V_x in the circuit of Fig. 13.144 using *PSpice* or *MultiSim*.

Figure 13.144



Step-by-step solution

Step 1 of 2

The schematic is shown below. In the AC sweep box, we set total pts = 1, Start frequency = 0.1592. After simulation, the output file which includes,

Frequency	$I\ M(V_Print1)$	$I\ P(V_Print1)$
$1.592\ E-01$	$1.080\ E+00$	$3.391\ E+01$
Frequency	$V\ M(\$N_0001)$	$V\ P(\$N_0001)$
$1.592\ E-01$	$1.514\ E+01$	$-3.421\ E+01$

i.e.,

$$i_x = 1.08 \angle 33.91^\circ\ A,$$

$$V_x = 15.14 \angle -34.21^\circ\ A.$$

Step 2 of 2

